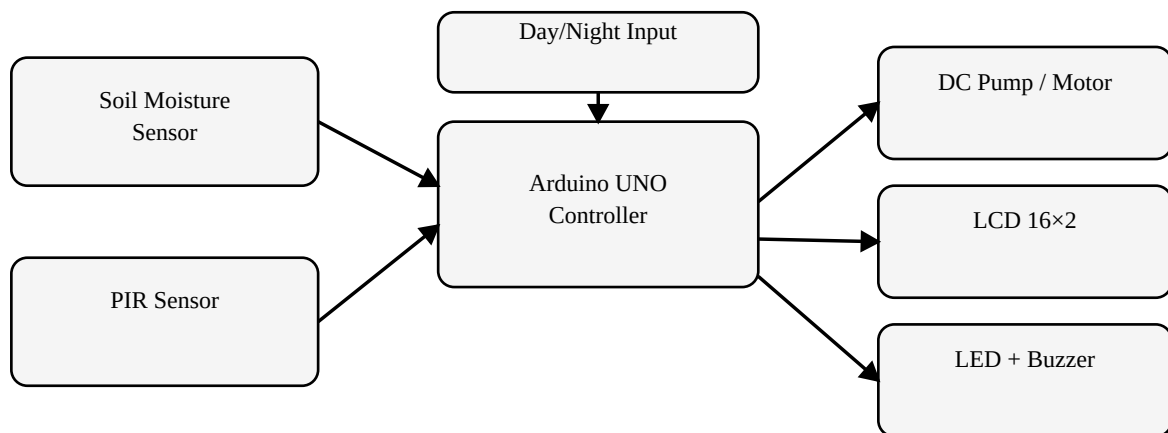


SMART IRRIGATION AND ANIMAL DETECTION SYSTEM

Arduino Uno Based Agriculture Monitoring and Automation Project

Project Report

This report presents an automated irrigation system combined with animal detection for agricultural fields. The system monitors soil moisture, controls a water pump, detects movement at night using a PIR sensor, and provides local alerts through an LED, buzzer, and LCD.



Prepared as an academic project report

1. Abstract

Agriculture depends on timely irrigation and protection of crops from environmental and biological threats. This project combines two useful functions in one Arduino-based system: **smart irrigation** and **animal detection**. A soil moisture sensor measures the moisture condition of the soil. When the measured value falls below a programmed threshold, the Arduino switches the DC pump ON. When sufficient moisture is present, the pump is switched OFF. A day/night input determines whether the animal-detection function is active. At night, a PIR sensor detects movement and the system activates an LED and buzzer while displaying an alert on the LCD.

The design demonstrates how low-cost embedded electronics can automate routine agricultural tasks, reduce unnecessary water use, and provide an early warning when movement is detected in a field. The system is intended as a prototype and can be expanded with wireless communication, additional sensors, solar power, and data logging.

Keywords: Arduino Uno, smart irrigation, soil moisture, PIR sensor, animal detection, agriculture, DC pump, LCD, automation.

2. Introduction

Traditional irrigation often depends on manual observation or fixed schedules. Both approaches can result in over-irrigation or under-irrigation because actual soil conditions change with weather, crop type, and field conditions. Automation allows irrigation decisions to be based on measured soil moisture rather than only on time.

Agricultural fields can also experience unwanted animal movement, particularly at night. A PIR sensor can provide a simple local indication of movement. In this project, the detection event is communicated using an LED, buzzer and LCD message. The system therefore combines water management with field monitoring in a single controller.

3. Problem Statement

Farmers may face two practical problems: inefficient irrigation and difficulty noticing animal movement during low-light periods. Manual irrigation requires time and may waste water, while continuous human monitoring of fields is difficult, especially at night.

The proposed project addresses these problems with a low-cost embedded system that senses soil moisture, controls a pump automatically, and detects movement at night.

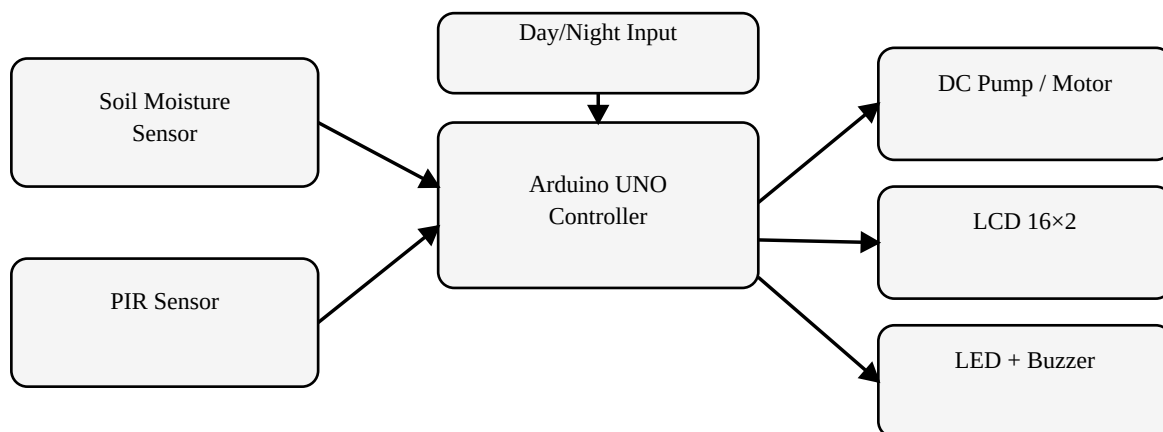
4. Objectives

- To monitor soil moisture continuously using a soil moisture sensor.
- To automatically switch the irrigation pump according to a moisture threshold.
- To detect movement in the agricultural area using a PIR sensor.
- To enable animal detection primarily during the night mode.
- To provide immediate local indication using an LED and buzzer.
- To display moisture and system status on a 16×2 LCD.
- To reduce unnecessary water usage through condition-based irrigation.
- To demonstrate an affordable and expandable agricultural automation prototype.

5. Why This Project Is Needed

- Water is a limited resource and should be supplied according to crop and soil requirements.
- Manual irrigation can require repeated field visits and supervision.
- Fixed-time irrigation cannot directly respond to changing soil moisture.
- Night-time movement in crop fields can go unnoticed without monitoring.
- A combined controller can reduce the number of separate systems required for basic field monitoring.
- The prototype can serve as a foundation for more advanced precision-agriculture systems.

6. Proposed System



The Arduino Uno acts as the central controller. It reads the soil moisture sensor through an analog input and reads the day/night control through another analog input. The PIR sensor provides a digital motion signal. Based on these inputs, the Arduino controls the motor/pump, LED, buzzer and LCD.

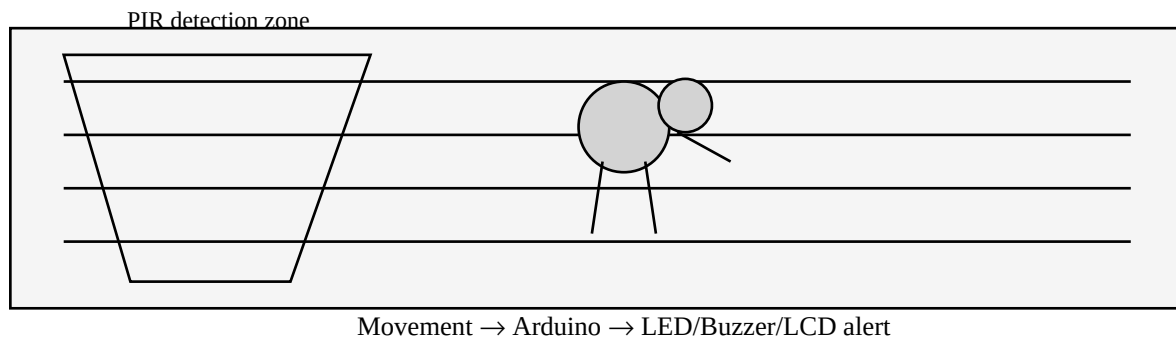
7. System Working Principle

The system begins by initializing the LCD, serial communication and output/input pins. During normal operation, the Arduino reads the soil moisture value and compares it with a threshold of 500 as specified in the provided program. If the moisture reading is below the threshold, the motor output is HIGH; otherwise it is LOW.

The day/night potentiometer provides a value used by the program to determine the mode. When the value is below 400, the system treats the condition as night. In night mode, the PIR input is checked. If motion is detected, the LED and buzzer are activated and the LCD displays an animal-detection alert. If there is no motion, the warning devices remain OFF.

During day mode, the PIR warning function is disabled in the provided logic. The LCD can show the current moisture and pump state under the normal-display condition.

8. Animal Detection Function



The animal detection part of the project uses a PIR (Passive Infrared) sensor. PIR sensors respond to changes in infrared radiation associated with moving warm objects. In this project, the sensor is used to detect movement in the monitored area rather than to identify the exact species of an animal.

When the system is in night mode and the PIR output becomes HIGH, the Arduino interprets this as motion. It then turns the LED ON, produces a buzzer tone, and displays “**NIGHT ALERT!**” and “**ANIMAL DETECTED**” on the LCD. This provides a local warning so that the farmer or operator can investigate the field.

Important limitation: PIR detection alone does not prove that the moving object is an animal. A person or another moving warm object may also trigger the sensor. For reliable species identification, the system would need additional sensing such as a camera or other classification technology.

9. Smart Irrigation Function

The soil moisture sensor is connected to an analog input. The Arduino obtains a numerical reading and compares it with the programmed threshold. The present code uses **500** as the threshold. If moisture is below this value, the pump is switched ON; otherwise it is switched OFF.

This condition-based approach is more responsive than simply running a pump for a fixed period. In a practical installation, the threshold should be calibrated for the specific sensor, soil, crop and growing conditions.

10. Components Used

Reference	Quantity	Component	Purpose
U1	1	Arduino Uno R3	Main controller
SEN1	1	Soil Moisture Sensor	Measures soil moisture
M1	1	DC Motor / Pump	Moves irrigation water
T1	1	NPN Transistor (BJT)	Switching driver for motor
D1	1	Diode	Suppresses inductive voltage from motor
BAT1	1	9V Battery	Power source shown in prototype
R1, R2	2	1 k Ω Resistors	Current limiting / circuit protection
U2	1	16 $\times$ 2 LCD	Displays system information
Rpot1, Rpot2	2	250 k Ω Potentiometers	User-adjustable analog inputs
PIR1	1	PIR Sensor	Detects movement
PIEZO1	1	Piezo Buzzer	Audible alert
D2	1	Red LED	Visual alert

11. Arduino Pin Configuration

Function	Arduino Pin	Type
Soil moisture sensor	A0	Analog input
Day/night control	A1	Analog input
Pump / motor control	D8	Digital output
PIR sensor	D10	Digital input
LED	D11	Digital output
Buzzer	D2	Digital output
LCD RS	D7	Digital output
LCD Enable	D6	Digital output
LCD D4	D5	Digital output
LCD D5	D4	Digital output
LCD D6	D3	Digital output
LCD D7	D9	Digital output

12. Software and Program Logic

The program is written for Arduino and uses the LiquidCrystal library for the 16×2 LCD. The main logic can be summarized as follows:

- Initialize pins and LCD.
- Read soil moisture from A0.
- Compare moisture with threshold 500.
- Turn the pump ON when moisture is below the threshold; otherwise turn it OFF.
- Read the day/night value from A1.
- Read motion from the PIR input on D10.
- If night and motion is detected, turn ON the LED and buzzer and display an animal-detection alert.
- If no motion is detected, turn OFF the LED and buzzer.
- In the normal LCD condition, display the moisture value and pump status.
- Repeat the loop continuously.

13. Agriculture Applications

- Automatic irrigation for vegetable gardens and small farms.
- Water management for nurseries and greenhouse plants.
- Moisture-based irrigation in home or educational agricultural setups.
- Night-time movement monitoring around crop fields.
- Prototype for precision-agriculture and smart-farming demonstrations.
- Integration with water tanks, pumps and additional environmental sensors.

14. How It Helps Agriculture

The system supports agriculture by reducing dependence on manual irrigation decisions and by making field monitoring more continuous. Moisture-based pump control can help avoid watering when the soil is already sufficiently moist. The animal-detection function adds a local warning layer for movement during night conditions.

The project is especially useful as an educational and prototype platform because the same Arduino controller can be expanded with temperature, humidity, rain, water-level, light, GSM/Wi-Fi or camera modules.

15. Advantages

- Automatic pump control based on soil condition.
- Can reduce unnecessary irrigation and save water when correctly calibrated.
- Low-cost and relatively simple hardware.
- LCD provides direct system information.
- Night-time movement detection provides an immediate local alert.
- Can operate without continuous internet connectivity.
- Arduino platform makes the system easy to modify and expand.
- Combines irrigation and field monitoring in one prototype.

16. Disadvantages and Limitations

- Soil moisture sensors can require calibration and may be affected by soil type and sensor condition.
- A PIR sensor detects movement but cannot reliably identify an animal species.
- False alerts may occur because people or other moving warm objects can trigger PIR detection.
- The 9V battery shown in the prototype may not be suitable for powering a real irrigation pump for long periods.
- A practical pump requires an appropriately rated driver, power supply, protection and wiring.
- The prototype provides local alerts; it does not automatically send a remote notification.
- Threshold values are fixed in the current program and may need adjustment for different crops and soils.

17. Safety and Practical Considerations

- Keep water away from exposed electronics and use suitable enclosures.
- Use a properly rated motor driver rather than relying directly on an Arduino output pin.
- Use a flyback diode across a DC motor/relay coil where appropriate.
- Provide a separate, adequately rated power supply for the pump.
- Protect wiring from moisture, corrosion and accidental short circuits.
- Calibrate the moisture threshold before real agricultural deployment.

18. Testing Plan and Expected Results

A prototype can be tested under several controlled conditions:

- Dry soil: moisture value should fall below the selected threshold and the pump should switch ON.
- Moist soil: moisture value should remain above the threshold and the pump should switch OFF.
- Night + motion: PIR should become HIGH and the LED, buzzer and LCD alert should activate.
- Night + no motion: LED and buzzer should remain OFF.
- Day + motion: the current program should keep the night PIR alert inactive.
- LCD test: moisture value and pump state should be visible in the normal display condition.

19. Future Scope

- Add GSM, Wi-Fi or LoRa communication for remote notifications.
- Add a water-level sensor to prevent the pump from running when the tank is empty.
- Add rain and temperature/humidity sensors for more intelligent irrigation.
- Replace the fixed threshold with adaptive or crop-specific control.
- Use a solar panel and rechargeable battery for off-grid operation.
- Add data logging and a mobile/web dashboard.
- Use a camera and computer vision if actual animal identification is required.
- Use multiple PIR sensors to cover larger fields.
- Add automatic scheduling and separate irrigation zones.

20. Conclusion

The Smart Irrigation and Animal Detection System demonstrates how an Arduino Uno can combine sensing, decision-making and actuator control for agricultural applications. The soil moisture sensor enables condition-based irrigation, while the PIR sensor provides movement detection during night mode. The LCD, LED and buzzer provide clear local feedback.

The project is a practical educational prototype for smart farming. With proper calibration, suitable power electronics and additional communication and sensing modules, the concept can be developed into a more capable agricultural monitoring and automation system.

21. References / Learning Resources

- Arduino Uno documentation and Arduino programming references.
- LiquidCrystal library documentation for 16×2 character LCDs.
- General datasheets/application notes for soil moisture sensors, PIR sensors, LEDs, buzzers, transistors and diodes.
- Basic principles of agricultural irrigation and soil-moisture-based watering.

Project title used throughout this revised report: Smart Irrigation and Animal Detection System